



Project Proposal

CSC-225 – Software Engineering
Department of Computer Science
Namal University, Mianwali

Project Title: Namal Criko

Team Members:

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Instructor: Asiya Batool

Semester: Fall 2025

Requirement Provider Agreement

This agreement is made between the student team and the Requirement Provider (RP) to ensure collaboration between them during the development of the project.

Requirement Provider Details:

Name: Mr.. Ammar Ahmad Khan

Designation: Professor Namal University Mianwali

Email: ammar.ahmad@namal.edu.pk

Student Representative Details:

Name: M.Jamal Ahmad Khan

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Agreement Terms:

- The RP agrees to act as a client throughout the duration of the project.
- The student team agrees to communicate regularly and seek RP's feedback.
- Meetings will occur once every two weeks, depending on the availability of both parties (virtual/physical).
- All discussions will be recorded in the Meeting Minutes document.

Date: 3rd November 2025



Student Representative



Requirement Provider

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1. Introduction

Cricket is the most popular sport played all over the world. It requires physical skills, continuous practice, and an awareness of techniques. Traditional training systems require a physical setup, coaches, and some other resources that often limit players to practice effectively.

With the advancement of technology, the sports training system has been changed to simulation and other tools. The purpose of this project, **Namal Criko**, is to provide a platform where players can learn different types of batting and bowling techniques by simulations.

Through this, players can experience different styles of bowling such as spin, fast, and slow deliveries, and also batting techniques (drive, sweep, and defensive shots, etc.) in a simulated environment. The main goal of this system is to provide knowledge of cricket techniques in a controlled and engaging environment. The system will provide training aid, especially to students and cricket lovers at **Namal University** who wish to improve their cricket knowledge and techniques using technology.

2. Problem Statement

In Namal, many students play cricket regularly, yet there isn't any proper system or structured guide to help them learn or practice the correct techniques of batting, bowling, and fielding. Most players rely on personal experience or random practice, which limits their learning. As of now, there is no digital or visual platform that explains how each technique should actually be performed. Because of this, learners find it difficult to understand the correct movements, timing, and form. Therefore, there is a need for an application that can simulate different cricket techniques and make learning easier and more effective.

3. Project Objectives

The objectives of the project are:

1. A user-friendly app that would show how various cricket techniques are executed, such as batting and bowling.
2. To help learners understand the right body movements and timing through visual animations.
3. Allow users to select any given shot or delivery for instance, a sweep shot or a Yorker-and watch step-by-step how it is done.

4. Making the learning of cricket more interesting and fun with the help of computer-based simulations, rather than just reading or watching videos. s
5. To provide students at Namal with an easy way to learn professional cricket skills and techniques.
6. To connect technology with learning in sports and help users understand how cricket techniques actually work.

4. Stakeholder Identification

Stakeholder	Role / Relation to System
Requirement Provider	Acts as the client; provides requirements and feedback.
Players	They are end users responsible to perform system operation.
Administrator	Responsible for managing user accounts, maintaining system functionality, ensuring data security, and implementing necessary updates or modifications.
Developer	Design,develop,maintain system.Responsible for implementing new features and removing technical errors and improve system performance

5. Software Development Methodology

We have adopted the **Agile (Scrum)** methodology for the development of **Namal Criko**. By utilizing 2-week sprints, we ensure that the complex 3D web simulations and administrative authorization workflows are developed iteratively with continuous feedback from the Requirement Provider (RP).

Scrum-Based One-Year Development Plan (Dec 2025 – Nov 2026)

Phase / Sprint	Scrum-Aligned Description
Product Backlog Creation	Dec 2025: Collaborative sessions with the RP to define User Stories for the Player and Admin portals. Priority is given to defining the "Admin Approval" logic and 3D simulation sub-categories (Fast/Slow bowling).
Sprint 0: Setup	Late Dec – Early Jan: Initializing the web environment using React.js and Three.js . Setting up the Firebase Web SDK for authentication and designing the dual-dashboard UI wireframes in Figma.

Sprints 1–3: Core Web Simulations	Jan – Mar 2026: Developing the 3D browser environment. Implementing basic player meshes, batting/bowling animations in Three.js, and ensuring the "Simulation + Written Description" layout is functional.
Sprints 4–6: Advanced Mechanics	Apr – Jun 2026: Expansion of simulation categories to include Fast, Swing, and Spin bowling variations. Implementing advanced batting styles and refining ball physics for a realistic web experience.
Sprints 7–8: Admin & User Systems	Jul – Aug 2026: Critical Phase: Implementation of the Admin Authorization Module (Allow/Approve users). Development of the Player Performance Tracker (Run Rate, Matches) and the Admin-controlled Ad Management system.
Sprints 9–10: Web Optimization	Sep – Oct 2026: Optimizing 3D assets for low-latency web loading. Integrating the Help Section/Support Desk and conducting rigorous cross-browser testing (Chrome, Safari, Edge).
Final Sprint: Deployment	Nov 2026: Final User Acceptance Testing (UAT) focusing on the Admin workflow and simulation accuracy. Deployment of the web application via Firebase Hosting or Vercel.

6. Tools and Technologies

For the development of **Namal Criko**, which is a web-based cricket training and simulation application, the following tools and technologies will be used. These tools are selected based on the requirements of 3D web rendering, browser performance, and secure administrative backend support.

- **Web Framework: React.js**

React.js will be used to develop the core web application, providing a responsive and component-based architecture for the separate Player and Admin dashboards.

- **Web Simulation Engine: Three.js (WebGL)**

To facilitate 3D simulations within a browser environment, Three.js will be utilized. This allows for the rendering of complex ball trajectories and player mechanics without requiring users to download additional software.

- **3D Modeling and Animation: Blender**

Blender will be used to design and animate 3D cricket assets, including players,

stadiums, and equipment. These models will be exported in glTF/GLB formats for optimized web loading.

- **Backend and Database: Node.js and MongoDB**

The backend will handle the Admin authorization logic, storage of player performance metrics (Run Rate, Matches), and the help-desk ticketing system.

- **UI/UX Design: Figma**

Figma will be used to design the web interfaces, ensuring a seamless user experience for both the learning modules and the administrative control panels.

- **Version Control: GitHub**

GitHub will be used for source code management, collaborative development, and maintaining project versions throughout the semester.

- **Project Management: Zoom / Google Sheets**

These tools will be used for coordinating team meetings and tracking task progress under the Agile Scrum methodology.

7. References

1. B. V. de Carvalho and C. H. P. Mello, “Scrum agile product development method – literature review, analysis and classification,” *Product: Management & Development*, vol. 9, no. 1, pp. 39–46, Jun. 2011.
2. N. Herdika and E. K. Budiardjo, “Variability and commonality requirement specification on agile software development: Scrum, XP, Lean, and Kanban,” in *Proc. 3rd Int. Conf. on Computer and Informatics Engineering (IC2IE)*, 2020, pp. 323–329.
3. “Firebase Documentation,” Google Firebase. [Online]. Available: <https://firebase.google.com/>

AI Usage Log

Prompts and tools used for idea generation:

- Chat GPT query: “Write a sample software engineering project proposal template in LaTeX along with a RP student General Agreement”
- Agreement sample generation
- Google Gemini for grammar and spell check
- Chat GPT for tools selection.